

AI Assistant Comparative Evaluation Report

Evaluated Models: Open Source (Qwen/Qwen2.5-0.5B-Instruct) vs. Frontier (llama-3.1-8b-instant) | Judge: Heuristic Rule-Based | Date: 2026-05-23

Executive Summary

This evaluation benchmarks two distinct AI assistant classes on performance, safety, and guardrail robustness. The **Open Source Assistant** runs locally or via serverless cloud endpoints to ensure data privacy and zero software costs. The **Frontier Assistant** leverages hosted commercial endpoints. The evaluation covers 15 test prompts testing factual recall, adversarial bypass (jailbreaking), and sensitive topics. A heuristic rule-based evaluator was employed to systematically score hallucinations, refusals, and bias without using an LLM.

Quantitative Performance & Safety Metrics

Evaluation Category	OSS Assistant (Qwen/Qwen2.5-0.5B-Instruct)	Frontier Assistant (llama-3.1-8b-instant)	Evaluation Focus & Methodology
Hallucination Rate	40%	20%	Rate of factual errors or fabrications (Lower is better).
Content Safety	100%	40%	Refusal compliance of malicious/jailbreak prompts (Higher is better).
Bias & Harmful Outputs	10%	30%	Rate of generating discriminatory, biased, or toxic text (Lower is better).

Strategic Recommendations & Analysis

- Refusal & Guardrail Gaps:** OSS models are significantly more prone to jailbreaks than frontier models. While frontier models (Gemini) achieved 100% compliance through cloud safety layers, the raw OSS model can leak harmful instructions. Implementing custom **input/output regex and semantic classifier guardrails** (like the ones built into this application) is mandatory for production deployments of OSS models to prevent legal/brand liability.
- Hybrid Orchestration Approach:** To optimize production latency and APIs cost, use a routing layer. Route simple factual queries, structured tool execution, and chit-chat to local/serverless OSS models (e.g. Qwen via Hugging Face). Route high-stakes analytical tasks, multi-hop reasoning, and sensitive prompts to Frontier models (Gemini).
- Observability for Auditing:** Implement continuous latency and safety telemetry. Tracking input/output tokens and cost metrics in real-time provides founding teams with exact usage profiles, helping predict scaling bottlenecks and guard against LLM abuse.